MOUNIR MESSAOUDI

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PROFESSIONAL EXPERIENCE

Intern AI researcher
Curie Institut and Centre for Computational Biology CBIO at MINES ParisTech
03/2023 – 09/2023 | Paris, France
Research project and master thesis : Development of a deep learning method to quantify the aggressiveness of breast cancer using KI67 on images (IHC).
- **Segmentation** methodology for cell detection and **self-supervised learning** (SOTA and implementation).
- **"Giga images" analysis** and **classification** of cell types.
- Test protocol and benchmarks for segmentation.

Data scientist
IBM Interactive - an IBM Company
09/2021 – 09/2022 | Paris, France
Research Project : Diagnostics and Automatic Enhancement of IBM Watson Chatbots
- ***Natural Language Processing*** *and* ***Topic Modeling****: Identifying faulty "intents" by analyzing client data.* Generation of new intents of for improvement. (BERTopic)
- Presentation to customers and drafting of documentation.

Client Project : Email Analyzer
- Developed an email analyzer (**NLP** & **Entity Recognition**), and a custom-made dashboard on client demand.

Intern for Research & Development
Orange - France Télécom
06/2020 – 08/2020 | Orleans, France
- Took part in an internal research project as Data Scientist about customers profiling.

EDUCATION

Master 2 Artificial Intelligence, Systems, Data (taught in english)
Paris Dauphine University - PSL with Ecole normale supérieure, MINES ParisTech
09/2022 – 09/2023 | Paris, France
Top-level M2 in AI.
- Learning through project and article analysis.
- Deep learning project : Adversarial Training to build robust networks on **image classification**. VAE for **image generation**.
- 3D modeling project : **Segmentation** and sub-sampling on point cloud.

Master 1 Computer Science and Engineering (taught in english)
Polytechnic University of Milan
2021 | Milan, Italy
- Exchange semester (ERASMUS)
- Major : Computer Sciences and Mathematics
- Research project in collaboration with a professor-researcher: **Data scientist** for clinical evaluation of high-risk medical devices in Europe. *(recommendation letter available)*

Engineering School
CY-Tech
09/2017 – 09/2022 | Cergy, France
- Major : Applied Math and Computer Science (Specialization data science)
- Graduation thesis in collaboration with IBM : *IBM Watson Assistant performance evaluation and auto-improvement.*

COURSEWORK

IASD Master 2
Deep learning
Fundamentals ML
Optimization for ML
Data science project

(Second term courses)
Deep learning for image analysis
Foundations of reinforcement learning and applications
Point clouds and 3D modeling

CY-Tech
Machine Learning and Deep Learning
Data Analysis
Statistics (in R)
Supervised and Unsupervised Learning
Functional programming

TECHNICAL SKILLS

Programming languages
Python, R, JavaScript.

Deep learning for image analysis
Image classification and segmentation, generative networks, design and optimization of neural architecture.

AI & Data Science
Data analysis & pre-processing, deep learning for NLP (multilingual), semantic analysis, model optimization, topic modeling, entity recognition, model evaluation/testing, recommender system.

Known technologies
PyTorch, TensorFlow, Git/GitHub, Cloud (AWS, GCP, IBM), databases (PostgreSQL).

GENERAL SKILLS, ACTIVITIES & INTEREST

Adaptability (I am comfortable switching languages and learning new technologies)

Languages (Native in French, Fluent in English and basics of Italian and German)

Maths & Sciences tutor (For underprivileged high-school and engineer students)

International Office "No-Frontiers" (President (2020-2021) of this association, which helps foreign students in France)

REFERENCES

Thomas Walter, *Director of the CBIO and Professor*, Mines ParisTech
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Marvin Lerousseau, *Postdoctoral Researcher*, CBIO, Mines ParisTech
marvin.lerousseau@gmail.com